Question 4 a.)

$\det(A - \lambda I) = 0$ $\quad -12 - 4 - 3$

$$A - \lambda I = \begin{pmatrix} -2-\lambda & 2 & -3 \\ 2 & 1-\lambda & -6 \\ -1 & -2 & 0-\lambda \end{pmatrix} \begin{matrix} -2-\lambda & 2 \\ 2 & 1-\lambda \\ -1 & -2 \end{matrix}$$

$$\left[ -(2+\lambda)(-\lambda)(1-\lambda) \pm 12 + 12 \right]$$

$$-\left[ 3(1-\lambda) - 12(2+\lambda) + 4\lambda \right] = 0$$

$$(\lambda - \lambda^2)(2+\lambda) + 24 - 3 + 3\lambda + 24 + 12\lambda + 4\lambda = 0$$

$$2\lambda + \lambda^2 - 2\lambda^2 - \lambda^3 + 24 - 3 + 3\lambda + 24 + 12\lambda + 4\lambda = 0$$

$$-\lambda^3 - \lambda^2 + 24\lambda + 45 = 0$$

If I plug in, let say 1 solution $\lambda = 5$ Poly $= 0$

$\lambda = -6 \pm \sqrt{\quad}$

$-\lambda^3 - \lambda^2 + 21\lambda + 45 = 0$

$\lambda = 5$, Pol $= 0$

$\lambda - 5 \;\big|\; \lambda^2 + 6\lambda + 9$

$-\lambda^3 + \lambda^2 - 21\lambda - 45$

$-\lambda^3 + 5\lambda^2$

$6\lambda^2 - 21\lambda - 45$

$-6\lambda^2 + 30\lambda - 45$

$9\lambda - 45$

$-9\lambda + 45$

$0$

$$\lambda - 5\left(\frac{(\lambda^3+\lambda^2-21\lambda-45)}{\lambda-5}\right) = 0$$

~~$(\lambda-5)(\lambda^3+\lambda^2-21\lambda$~~

$$(\lambda-5)(\lambda^2+6\lambda+9) = 0$$

$$\lambda = \frac{-6 \pm \sqrt{36-36}}{2(1)}$$

$$\lambda_{2,3} = -3$$

$$\lambda_1 = 5$$